

Housing Allocation and Fertility

An illustrative two-period model

Discussion draft for Tommaso De Santo September 10, 2026

Young households need space for children, but may be unable to finance a large home. Renting provides an alternative only up to the largest rental available. We ask whether these two restrictions leave too little housing with the young, and whether reallocating housing can increase fertility.

The comparison is between a competitive allocation and a planner who chooses all current consumption and housing, giving each living household equal weight. The planner can relax private financing restrictions. We first hold fertility fixed, solve for the entire allocation, and give sufficient conditions for housing to move toward the young. We then allow fertility to respond. Both results permit the planner to give the young more housing while taking some consumption from them. The direction of the housing transfer requires a restriction on relative resources; having children alone does not establish it.¹

1 Environment

Households. Households live for two periods, young and old. A young household has resources $w > 0$, distributed according to F , and receives a common income $y > 0$ upon entering old age. Resources w include income and liquid wealth; all are available for consumption, housing, and saving. The same distribution F enters each cohort. Future income is known. Resource and utility averages are assumed finite.

Preferences. A young household chooses consumption expenditure c^y , housing h^y , and completed fertility $n > 0$. Children require $\chi > 0$ units of goods and $\kappa > 0$ units of housing each. Utility at the two ages is:

$$u^y(c^y, h^y, n) = \log(c^y - \chi n) + \alpha \log(h^y - \kappa n) + \vartheta \log n, \quad (1)$$

$$u^o(c^o, h^o, e) = \log c^o + \alpha \log h^o + \omega_B \log e. \quad (2)$$

Here $\alpha, \vartheta, \omega_B > 0$, and the arguments of all logarithms are positive. Households discount future utility at $\beta > 0$. The estate e is financial wealth plus the proceeds from selling the house at death, net of mortgage repayment. Parents value this estate directly. Entering households' resources are specified independently of the estates left by their parents.

Housing. A fixed, divisible housing stock $\bar{H} > 0$ supplies both owned and rented homes. Rentals have maximum size $r > 0$; owned homes can have any size. Households choose tenure at each age. Converting between the two uses does not consume resources, so the stock is common rather than divided into separately fixed owner and rental stocks.

Finance and taxes. Goods and financial claims trade with the rest of the world. The price of a claim to one unit of goods at the end of an age is $q \in (0, 1)$. At stationary prices, a unit of

¹This draft develops the proposed model with interest paid during each age, a common positive retirement income, and the same adult housing weight at both ages. Housing comes from a common stock, with larger units available through ownership. Property taxes finance public services; entering households' endowments are independent of estates. These assumptions should be assessed together with the results. Robustness beyond logarithms, additive separability, and fixed per-child requirements remains to be assessed.

housing sells for $P > 0$ and owners pay property tax τP , with $\tau > 0$. Competitive landlords charge current rent:

$$p = (1 + \tau - q)P. \quad (3)$$

Rent covers the purchase and tax cost less the discounted resale proceeds. The return $1/q$ applies over one model period. Property-tax revenue finances public services; their provision is held fixed in the allocation comparison.

A mortgage finances at most a share $\phi \in (0, 1)$ of the purchase price. The borrower pays interest during the age and repays principal when selling the house at its end. The same secured borrowing rule applies at both ages. Financial saving is nonnegative; borrowing requires housing collateral. Old households can sell their previous home and choose its replacement.

Demography. There are Y_t young and O_t old households. Each child produces $\nu > 0$ entering households after survival and household formation. If $\bar{n}_t$ is mean fertility, cohort masses evolve as:

$$Y_{t+1} = \nu \bar{n}_t Y_t, \quad O_{t+1} = Y_t. \quad (4)$$

The stationary comparison below has equal young and old cohort masses.

2 Household choices

2.1 Budgets and mortgage payments

Let z be total resources upon entering old age, including retirement income. For an owner, let L be mortgage principal and k financial saving. Since principal L is repaid at sale, the current value of its interest payments is $(1 - q)L$. The purchase and current expenditure budget is:

$$c + (1 + \tau)Ph + k + (1 - q)L = w + L, \quad k \geq 0, \quad 0 \leq L \leq \phi Ph. \quad (5)$$

The household finances the deposit, consumption, tax, and interest from its current resources. This is an interest-serviced mortgage, rather than a mortgage whose interest is added to principal until retirement.

Define the net financial position $a = k - qL$. The owner's current budget and retirement resources become:

$$\begin{aligned} c + (1 + \tau)Ph + a &= w, \\ z &= y + a/q + Ph, \\ a &\geq -q\phi Ph. \end{aligned} \quad (6)$$

The house sale repays the unchanged mortgage principal and releases the remaining equity. A renter instead has $c + ph + a = w$ and $z = y + a/q$, with $a \geq 0$ and $h \leq r$.

Eliminating a gives the same lifetime budget in both tenures:

$$c + ph + qz = w + qy. \quad (7)$$

The difference is which combinations of housing and retirement resources are available. Define the positive coefficient $\Gamma = q(1 - \phi)/(1 + \tau - q)$. The two financing restrictions are:

$$\begin{aligned} \text{Renting: } z &\geq y, & h &\leq r, \\ \text{Owning: } q(z - y) &\geq \Gamma ph. \end{aligned} \quad (8)$$

For an owner, the last condition is equivalently $z - y \geq (1 - \phi)Ph$: the equity left in the house at sale must cover the unfinanced share of its price. Retirement income can pay for old-age consumption but cannot be pledged as unsecured borrowing when young.

2.2 The old household's problem

An old household enters with resources z^o , after receiving income and settling its previous financial and housing positions. It chooses current consumption, housing, and its net estate. Its budget is:

$$c^o + ph^o + qe = z^o. \quad (9)$$

For an owner, the estate satisfies $e = a^o/q + Ph^o$ with $a^o \geq -q\phi Ph^o$; for a renter, $e = a^o/q$ with $a^o \geq 0$. Thus the old household solves:

$$\begin{aligned} V^o(z^o) = \max_{c^o, h^o, e} \quad & \log c^o + \alpha \log h^o + \omega_B \log e, \\ \text{subject to} \quad & c^o + ph^o + qe = z^o, \\ & \{h^o \leq r, e \geq 0\} \quad \text{or} \quad \{qe \geq \Gamma ph^o\}. \end{aligned} \quad (10)$$

The alternatives are renting and owning. Estate utility does not describe consumption after death: it is the household's valuation, while alive, of what it leaves.

Throughout the results, assume:

$$\omega_B \geq \alpha \frac{q(1-\phi)}{1+\tau-q}. \quad (\text{A1})$$

Under this condition the old household's desired estate is large enough to satisfy the equity requirement on its desired home. Its solution is:

$$c^o = \frac{z^o}{1+\alpha+\omega_B}, \quad h^o = \frac{\alpha z^o}{(1+\alpha+\omega_B)p}, \quad qe = \frac{\omega_B z^o}{1+\alpha+\omega_B}. \quad (11)$$

If the home exceeds r , ownership implements this allocation. Otherwise renting does as well. The housing choice is therefore unrestricted in real terms, and $V^o(z) = (1+\alpha+\omega_B) \log z$ plus a term depending only on prices. Equality in (A1) is allowed.

2.3 The young household's problem

A young household chooses fertility, current consumption and housing, and resources for retirement, taking prices as given:

$$\begin{aligned} \max_{c, h, n, z} \quad & \log(c - \chi n) + \alpha \log(h - \kappa n) + \vartheta \log n + \beta V^o(z), \\ \text{subject to} \quad & c + ph + qz = w + qy, \\ & \{z \geq y, h \leq r\} \quad \text{or} \quad \{q(z - y) \geq \Gamma ph\}. \end{aligned} \quad (12)$$

The household compares the best rental and ownership plans, including their fertility and saving choices. The results do not prescribe which young households rent or own.

3 Stationary equilibrium

A stationary equilibrium links the housing choices of both ages to the same stock. Old households' resources are generated by their choices when young. For any variable indexed by entering type, a bar denotes its average under F .

Definition 1. A positive stationary competitive equilibrium consists of prices P, p , cohort mass $N > 0$, and household choices $\{c^y(w), h^y(w), n(w), z(w), c^o(w), h^o(w), e(w)\}$ and tenures, such that:

1. Young households solve (12), and old households solve (10) with resources $z^o(w) = z(w)$ inherited from the preceding cohort's choices.
2. Landlords satisfy (3) and the housing market clears:

$$N(\bar{h}^y + \bar{h}^o) = \bar{H}. \quad (13)$$

3. Fertility replaces the entering cohort, $\nu\bar{n} = 1$, and public-service spending equals property-tax revenue $\tau P\bar{H}$.

Financial claims are settled with the external financial sector. Every young plan has $z \geq y$, so the resulting old resource distribution also satisfies $z^o \geq y$.

We take such an equilibrium as the allocation to be assessed. The propositions below hold at any stationary equilibrium satisfying their stated inequalities. The rental price p in those inequalities is an equilibrium object. The results do not rely on a calibrated reference equilibrium, but also do not assert global existence or uniqueness for arbitrary primitives.

4 The planner's allocation at fixed fertility

Would a planner give the young more housing? Hold each parent's fertility at its competitive choice n_i . The planner chooses all current consumption and housing, keeping each young household's retirement resources z_i and each old household's net estate e_j fixed. It can relax private finance and reassign tenure; a home larger than r must be owned. Total goods and housing remain those of the competitive allocation.

Let $\mathcal{C} = \bar{c}^{y,E} + \bar{c}^{o,E}$ and $\mathcal{H} = \bar{h}^{y,E} + \bar{h}^{o,E} = \bar{H}/N$ be these totals per young household, including both ages. Here E denotes the fixed competitive choices, including $n^E(w)$, $z^E(w)$, and $e^E(w)$. The planner's objective counts the full remaining utility of the N young and N old households, while the resource constraints are written per young household. Because future resources, continuation prices, and estates are fixed, the terms involving them do not affect the current allocation, but they remain in the objective to make clear whose welfare is counted. The planner's problem is:

$$\begin{aligned} \max_{\{c^y(w), h^y(w), c^o(w), h^o(w)\}} \quad & N \int \left[u^y(c^y(w), h^y(w), n^E(w)) + \beta V^o(z^E(w)) \right. \\ & \left. + u^o(c^o(w), h^o(w), e^E(w)) \right] dF(w), \\ \text{subject to} \quad & \int [c^y(w) + c^o(w)] dF(w) = \mathcal{C}, \\ & \int [h^y(w) + h^o(w)] dF(w) = \mathcal{H}. \end{aligned} \quad (14)$$

All adult consumption and space must be positive. This compares allocations for the living households at one date, including the initial old.

The planner divides adult goods and space equally and then adds children's needs. In particular, define:

$$X = \frac{\mathcal{C} - \chi\bar{n}}{2}, \quad S = \frac{\mathcal{H} - \kappa\bar{n}}{2}. \quad (15)$$

Strict concavity gives the unique real allocation:

$$\begin{aligned} (c_i^{y,F}, h_i^{y,F}) &= (X + \chi n_i, S + \kappa n_i), \\ (c_j^{o,F}, h_j^{o,F}) &= (X, S). \end{aligned} \tag{16}$$

Each parent receives κn_i more housing than an old household in this allocation. To compare with equilibrium, however, we must establish which age initially has more space available to adults.

4.1 When housing moves toward the young

Consider young types satisfying these two inequalities:

$$w < \frac{(1 + \alpha + \vartheta)qy}{\beta(1 + \alpha + \omega_B)}, \quad \frac{w}{1 + \alpha + \vartheta} \left(\frac{\alpha}{p} + \frac{\vartheta\kappa}{\chi + \kappa p} \right) > r. \tag{A2}$$

The first says the household would like to borrow against retirement income if it could rent a home of any size. The second says its preferred rental, subject to nonnegative saving, would exceed the rental ceiling. Ownership is available, but requires it to carry positive equity into retirement. Appendix A shows that these inequalities imply a strict housing distortion whichever tenure the household chooses. They are a simple sufficient test; the appendix gives the wider exact income interval.

The last condition compares the resources of the two ages:

$$\frac{y}{1 + \alpha + \omega_B} \geq \frac{\bar{w}}{1 + \alpha + \vartheta}. \tag{A3}$$

The left side is a floor on old consumption, since every old household has resources at least y . The right side bounds young current expenditure per unit of current utility weight, before subtracting saving and equity.

Proposition 1 (Housing allocation). *Suppose (A1) holds and a positive share of young households satisfies (A2). The competitive allocation does not maximize the planner's objective (14). If, in addition, (A3) holds, its maximizing allocation gives the young more housing in aggregate:*

$$\bar{h}^{y,F} - \bar{h}^{y,E} = \frac{\bar{h}^{o,E} - \overline{h^{y,E} - \kappa n}}{2} > 0. \tag{17}$$

An individual young household receives more housing exactly when $h_i^{y,E} - \kappa n_i < S$.

The borrowing restriction and rental ceiling establish a difference in marginal housing values. Condition (A3) additionally makes the full equal-weight allocation move housing toward the young as a group. The distinction matters because the planner also redistributes resources across rich and poor households within each age.

Here is the argument. Let $x_i = c_i^{y,E} - \chi n_i$ be young adult goods, $s_i = h_i^{y,E} - \kappa n_i$ adult space, and $B = \bar{c}^{o,E}$ mean old consumption. Define $v_i = ps_i/\alpha$ and the expenditure index $\hat{x}_i = (c_i^{y,E} + ph_i^{y,E})/(1 + \alpha + \vartheta)$. Household optimality gives:

$$v_i \leq \hat{x}_i \leq x_i, \quad \hat{x}_i = \frac{w_i - q(z_i - y)}{1 + \alpha + \vartheta} \leq \frac{w_i}{1 + \alpha + \vartheta}. \tag{18}$$

The first two inequalities are strict for the affected households. Old optimality gives $ph_j^{o,E}/\alpha = c_j^{o,E}$ and $B \geq y/(1+\alpha+\omega_B)$. Consequently, (A3) implies $B > \bar{v}$, which is exactly the housing comparison in (17).

This also shows a less restrictive sufficient condition than (A3): $B \geq \bar{x}$. The mean housing change can be written as:

$$\bar{h}^{y,F} - \bar{h}^{y,E} = \frac{\alpha}{2p} \left[(B - \bar{x}) + (\bar{x} - \bar{v}) \right]. \quad (19)$$

The first term compares the resources available to the two ages; the second comes from the young households' distorted division between goods and space.

Young consumption can fall even as housing rises. Its mean change is:

$$\bar{c}^{y,F} - \bar{c}^{y,E} = \frac{B - \bar{x}}{2}. \quad (20)$$

For example, at $B = \bar{x}$, the strict inequalities in (18) imply more young housing and less young consumption. The allocation corrects the balance between the two goods.

4.2 What the conditions require

Condition (A1) ensures that old housing demand has the usual ratio $h^o/c^o = \alpha/p$. It is a restriction on estate preferences relative to the required equity. If it fails, the old may also face a housing distortion and the comparison must be rederived.

At the unconstrained optimum, ω_B/α is the ratio $qe/(ph^o)$. As an illustration only, suppose an age lasts 20 years, the annual interest rate is 4%, the annual end-of-year property tax is 1%, and $\phi = .8$. Then $q = (1.04)^{-20} \approx .456$ and $\tau = .01 \sum_{j=1}^{20} (1.04)^{-j} \approx .136$, so the lower bound on ω_B/α is about .134. If $\alpha = .3$, the lower bound on ω_B is about .040, and at that bound the share of resources left as an estate, qe/z^o , is about 3%. These are illustrative inputs, not a calibration or empirical evidence about the mildness of the condition.

Condition (A2) identifies affected households from income, retirement resources, and access to rental space. It does not assume that they rent, or that they already have smaller homes than the old. Its first inequality is less likely to hold for patient households at given income. There is no imposed ordering between β and q , but this does not remove patience from the conditions.

Condition (A3) is sufficient rather than necessary. Written as an income ratio, it requires:

$$\frac{y}{\bar{w}} \geq \frac{1 + \alpha + \omega_B}{1 + \alpha + \vartheta}. \quad (21)$$

If $\omega_B \leq \vartheta$, then $y \geq \bar{w}$ suffices. The condition can be conservative because old resources include accumulated assets and young current expenditure excludes saving. It remains a substantive restriction, and these two-period endowments have not been mapped into annual income measures or calibration targets.

5 Fertility

5.1 The full allocation when fertility can change

Allow the planner to choose fertility as well as every consumption and housing bundle. It values existing parents' utility, including their taste for children. The set of living adults, individual

retirement resources, and net estates remains fixed. For any mean fertility, equal adult goods and space maximize utility; strict concavity of $\log n$ also makes fertility the same across parents. The full problem therefore reduces to:

$$\max_{0 < n < \min\{\mathcal{C}/\chi, \mathcal{H}/\kappa\}} 2 \log X(n) + 2\alpha \log S(n) + \vartheta \log n, \quad X(n) = \frac{\mathcal{C} - \chi n}{2}, \quad S(n) = \frac{\mathcal{H} - \kappa n}{2}. \quad (22)$$

The objective is strictly concave and approaches minus infinity at the feasible boundaries. Its unique solution n^F satisfies:

$$\frac{\vartheta}{n^F} = \frac{\chi}{X(n^F)} + \frac{\alpha \kappa}{S(n^F)}. \quad (23)$$

The marginal utility of another child equals the goods and housing it uses. Appendix C gives the explicit solution of this quadratic.

Every bundle at the optimum is then determined:

$$\boxed{\begin{aligned} (c^{y,F}, h^{y,F}) &= \left(\frac{\mathcal{C} + \chi n^F}{2}, \frac{\mathcal{H} + \kappa n^F}{2} \right), \\ (c^{o,F}, h^{o,F}) &= \left(\frac{\mathcal{C} - \chi n^F}{2}, \frac{\mathcal{H} - \kappa n^F}{2} \right). \end{aligned}} \quad (24)$$

The change in fertility changes the resources reserved for children as well as the resources shared by adults.

To sign the fertility comparison, assume:

$$\boxed{\kappa p \geq \alpha \chi}. \quad (A4)$$

Children's housing-to-goods expenditure ratio, $\kappa p / \chi$, must be at least the unrestricted adult ratio α . This condition concerns the composition of children's needs, rather than the size of the welfare gain.

Proposition 2 (Fertility). *Under the conditions for the aggregate housing result in Proposition 1 and (A4), the full planner chooses fertility above its competitive mean: $n^F > \bar{n}^E$. Aggregate housing assigned to the young also rises. Moreover, each parent voluntarily chooses n^F when given the fixed consumption and housing bundle in (24), with continuation resources held fixed.*

The private fertility condition helps explain the result. At any competitive bundle it is:

$$\frac{\vartheta}{n_i} = \frac{\chi}{x_i} + \frac{\kappa p}{v_i}. \quad (25)$$

Restricted housing makes space expensive at the margin. Under (A4), correcting the division of a parent's expenditure between goods and space increases fertility. Condition (A3) ensures that the full allocation has enough resources to preserve this gain in the aggregate, even though young consumption may fall. The proof uses (18) and the concavity of fertility as a function of adult goods and space.

The voluntary-choice statement uses the final bundles in (24). Give each parent those totals before it chooses fertility. At n^F , its residual adult bundle is $(X(n^F), S(n^F))$, so its private first-order condition is precisely (23). Strict concavity makes this its unique choice. The bundle need not be conditional on realized births.

This differs from first computing (16) at each parent's competitive fertility and then allowing parents to choose again. That sequence assigns different total bundles to parents with different initial fertility; it need not increase mean fertility. Proposition 2 instead computes the best allocation anticipating parents' choices.

5.2 A compensated housing transfer

The same argument has a local version that does not require the average income condition. For a household satisfying (A2), its marginal housing value in consumption units exceeds that of any old household:

$$\frac{u_h^y}{u_c^y} = \frac{\alpha x_i}{s_i} > p = \frac{\alpha c_j^o}{h_j^o} = \frac{u_h^o}{u_c^o}. \quad (26)$$

Transfer a little housing to that parent and enough goods in the opposite direction to compensate the old household. The parent gains after paying this compensation, even at its original fertility. Finance is relaxed as in the planner comparison, and retirement resources and estates are preserved.

If the parent subsequently chooses fertility at its new fixed bundle, the first-order fertility response is positive precisely when:

$$\kappa p \left(\frac{x_i}{v_i} \right)^2 > \alpha \chi. \quad (27)$$

Since $x_i/v_i > 1$ for the affected parent, the weak inequality (A4) suffices. The calculation is in Appendix D.

6 From allocation to policy and transition

The results establish a static allocation benchmark. They give the full planner allocation, sufficient conditions for housing to move toward the young, and a fertility comparison based on parents' own utility. The planner can assign fixed consumption and housing bundles while parents choose fertility at those bundles.

A policy using transfers while households freely choose housing, tenure, and saving is a different problem. Its effect depends on prices and on which constraints bind after the transfers. The direct allocation permits a relaxation of finance; the results above are therefore not a theorem of constrained inefficiency under the original borrowing limits.

Likewise, applying the fertility result to a dated allocation does not establish an equilibrium transition. To compare the paths following a fertility shock and a housing policy, one must specify that policy, solve for changing house prices and inherited wealth, and follow cohort entry. A new stationary population must satisfy replacement fertility, $\nu \bar{n} = 1$; its level also depends on housing per household through market clearing. Higher fertility at one allocation does not by itself rank those stationary population levels.

A Household choices and the affected incomes

A.1 Old households

Ignoring tenure temporarily, the old household allocates expenditure across three log arguments. The first-order conditions imply expenditure shares $1/(1+\alpha+\omega_B)$, $\alpha/(1+\alpha+\omega_B)$, and $\omega_B/(1+\alpha+\omega_B)$, giving (11). For that allocation, the owner restriction is:

$$qe - \Gamma ph^o = \frac{(\omega_B - \alpha\Gamma)z^o}{1 + \alpha + \omega_B} \geq 0. \quad (28)$$

Thus it is feasible under (A1), at any desired home size. It maximizes utility over the unrestricted budget set and hence over the union of the two tenure sets.

A.2 Young households

Write $x = c - \chi n$, $s = h - \kappa n$, $t = qz$, $Y = qy$, $\delta = \beta(1 + \alpha + \omega_B)$, and $C(p) = \chi + \kappa p$. Dropping terms constant in the choice variables, the young household's problem is:

$$\max\{\log x + \alpha \log s + \vartheta \log n + \delta \log t\}, \quad x + ps + C(p)n + t = w + Y. \quad (29)$$

Renting adds $t \geq Y$ and $s + \kappa n \leq r$; owning adds $t \geq Y + \Gamma p(s + \kappa n)$. Each conditional problem is strictly concave on a convex feasible set.

Let λ be the budget multiplier. If $\eta \geq 0$ is the rental-ceiling multiplier and $\mu \geq 0$ the owner's equity multiplier, the consumption and housing conditions are:

$$\lambda = 1/x, \quad \frac{\alpha}{s} = \begin{cases} p/x + \eta, & \text{renting,} \\ p/x + \Gamma p\mu, & \text{owning.} \end{cases} \quad (30)$$

Every optimal plan therefore has $\alpha x/s \geq p$. This remains true at a tie between tenure values.

Under the first inequality in (A2), the uncapped rental solution has $t = Y$, $x = w/(1 + \alpha + \vartheta)$, and:

$$s = \frac{\alpha w}{(1 + \alpha + \vartheta)p}, \quad n = \frac{\vartheta w}{(1 + \alpha + \vartheta)C(p)}, \quad h = \frac{w}{1 + \alpha + \vartheta} \left(\frac{\alpha}{p} + \frac{\vartheta \kappa}{C(p)} \right). \quad (31)$$

The second inequality in (A2) makes this home too large to rent, so the actual rental optimum has $\eta > 0$. For ownership, a zero multiplier would give the unrestricted lifetime solution $t = \delta(w + Y)/(1 + \alpha + \vartheta + \delta) < Y$, contradicting owner finance. Hence $\mu > 0$ at the owner optimum too. Either optimal tenure has a strict housing distortion.

For completeness, the exact affected-income interval can be calculated without solving for the tenure cutoff. Define:

$$A(p) = \alpha + \frac{\vartheta \kappa p}{C(p)}, \quad x_r(p) = \frac{pr}{A(p)}, \quad w_r(p) = (1 + \alpha + \vartheta)x_r(p) + \max\{\delta x_r(p) - Y, 0\}. \quad (32)$$

Here w_r is the income where the uncapped rental optimum first reaches r ; max denotes the larger of its two arguments. The unrestricted lifetime allocation can be financed through ownership once income reaches:

$$w_O(p) = \begin{cases} \frac{Y[1 + \alpha + \vartheta + \Gamma A(p)]}{\delta - \Gamma A(p)}, & \delta > \Gamma A(p), \\ +\infty, & \delta \leq \Gamma A(p). \end{cases} \quad (33)$$

Consequently, every globally optimal tenure plan has a strict housing distortion exactly for $w_r(p) < w < w_O(p)$.

To see both directions, below w_r the best uncapped rental is feasible. The owner's feasible set is contained in the uncapped rental set, so an owner cannot improve on it. At or above w_O , the fully unrestricted lifetime optimum is feasible through ownership. Between the thresholds, neither conditional optimum can have a zero housing multiplier. Conversely, a zero rental multiplier would make the uncapped rental feasible, and a zero owner multiplier would make the unrestricted lifetime optimum feasible. These statements include tenure ties and weak endpoint equalities. The interval is nonempty exactly when $\delta pr < A(p)[Y + \Gamma pr]$.

B The allocation comparison

B.1 Current expenditure and adult space

Retain x_i, s_i and $v_i = ps_i/\alpha$. Write $t_i = x_i/v_i \geq 1$ and $\lambda_c = \chi/(\kappa p) > 0$. The private fertility condition gives:

$$n_i = \frac{\vartheta v_i t_i}{\kappa p(\lambda_c + t_i)}, \quad \hat{x}_i = \frac{x_i + \alpha v_i + C(p)n_i}{1 + \alpha + \vartheta}. \quad (34)$$

Subtracting v_i and subtracting this index from x_i gives:

$$\hat{x}_i - v_i = \frac{v_i(t_i - 1)}{1 + \alpha + \vartheta} \left(1 + \frac{\vartheta \lambda_c}{\lambda_c + t_i} \right), \quad (35)$$

$$x_i - \hat{x}_i = \frac{v_i(t_i - 1)}{1 + \alpha + \vartheta} \left(\alpha + \frac{\vartheta t_i}{\lambda_c + t_i} \right). \quad (36)$$

Both expressions are nonnegative, and strictly positive when $t_i > 1$. This proves the first two inequalities in (18); its last inequality follows from $z_i \geq y$ and the budget.

B.2 The full maximizing allocation

At fixed fertility the planner's goods and housing first-order conditions respectively equalize $1/(c_i^y - \chi n_i)$ with $1/c_j^o$, and $\alpha/(h_i^y - \kappa n_i)$ with α/h_j^o , across all households. Resource feasibility gives (15). Strict concavity establishes the unique real optimum (16).

Since old housing is $\alpha c^o/p$, the adult allocations can also be written:

$$X = \frac{\bar{x} + B}{2}, \quad S = \frac{\alpha \bar{v} + B}{p \cdot 2}. \quad (37)$$

Condition (A3), (18), and a positive share with $t_i > 1$ imply $B \geq y/(1 + \alpha + \omega_B) \geq \bar{w}/(1 + \alpha + \vartheta) \geq \bar{\bar{x}} > \bar{v}$. Substitution proves the aggregate housing result and its individual and consumption comparisons.

B.3 Preserving retirement resources and estates

The allocation can be settled without changing the promised future resources. For a young household assigned an owned home h'_i , choose its net financial position as $a'_i = q(z_i - y) - qPh'_i$. For an old household with fixed estate e_j , use $a'_j = qe_j - qPh'_j$. Their current transfers are:

$$T_\ell = \Delta c_\ell + p\Delta h_\ell, \quad \bar{T}^y + \bar{T}^o = 0. \quad (38)$$

The transfers balance because total current consumption and housing are fixed. Allowing mortgages up to the full house value makes these financial positions feasible: $a'_i \geq -qPh'_i$ and $a'_j > -qPh'_j$ since $z_i \geq y$ and $e_j > 0$. Rental assignments use the corresponding nonnegative saving positions. Intermediaries adjust their counterpart positions, and the tax base $P\bar{H}$ is unchanged. This verifies feasibility for a planner allowed to relax finance; it does not implement the allocation under the original mortgage limit and unrestricted household reoptimization.

C Fertility and the explicit optimum

C.1 The aggregate fertility comparison

Define $g(x, v) = (\chi/x + \kappa p/v)^{-1}$, so private fertility is $n_i = \vartheta g(x_i, v_i)$. Direct substitution in (34) yields:

$$\frac{\hat{x}_i}{C(p)} - g(x_i, v_i) = \frac{v_i(t_i - 1)(t_i - \alpha\lambda_c)}{(1 + \alpha + \vartheta)C(p)(\lambda_c + t_i)}. \quad (39)$$

Under (A4), $\alpha\lambda_c \leq 1$, so this expression is nonnegative for every type and strictly positive for affected types. Thus $B/C(p) \geq \bar{x}/C(p) > \bar{n}/\vartheta$.

The function g is increasing and concave on positive adult bundles. Its second differential is:

$$d^2g = -\frac{2\chi\kappa p(v dx - x dv)^2}{(\chi v + \kappa p x)^3} \leq 0. \quad (40)$$

At the fixed-fertility planner allocation, let $V = pS/\alpha = (\bar{v} + B)/2$. Concavity then implies:

$$\begin{aligned} g(X, V) &\geq \frac{g(\bar{x}, \bar{v}) + g(B, B)}{2} \\ &\geq \frac{g(x_i, v_i) + B/C(p)}{2} > \frac{\bar{n}}{\vartheta}. \end{aligned} \quad (41)$$

At $n = \bar{n}$, the derivative of (22) is therefore $\vartheta/\bar{n} - 1/g(X, V) > 0$. Strict concavity proves $n^F > \bar{n}$. Jointly changing fertility adds $\kappa(n^F - \bar{n})/2$ to mean young housing relative to the fixed-fertility allocation. It adds $\chi(n^F - \bar{n})/2$ to mean young consumption, so a consumption decline in the first comparison need not persist in the joint optimum.

C.2 Closed-form solution

Multiplying (23) by its positive denominators gives a quadratic. With $D = \vartheta + 2(1 + \alpha)$ and $B = \kappa C(\vartheta + 2\alpha) + \chi \mathcal{H}(\vartheta + 2)$, it is:

$$\chi \kappa D n^2 - B n + \vartheta \mathcal{C} \mathcal{H} = 0. \quad (42)$$

The smaller root is the unique feasible optimum:

$$n^F = \frac{2\vartheta \mathcal{C} \mathcal{H}}{B + \sqrt{B^2 - 4\chi \kappa D \vartheta \mathcal{C} \mathcal{H}}}. \quad (43)$$

Its discriminant is strictly positive because it equals:

$$[\kappa C(\vartheta + 2\alpha) - \chi \mathcal{H}(\vartheta + 2)]^2 + 16\alpha \chi \kappa \mathcal{C} \mathcal{H}. \quad (44)$$

The derivative in (23) decreases from positive to negative infinity over the feasible interval, proving that exactly one root lies there; the quadratic is positive at zero, so that root is the smaller one. This is a closed-form solution for the planner conditional on the resource totals. It is not a claim that every tenure choice and competitive price has an elementary closed form.

C.3 Voluntary fertility at the final bundles

Give every parent the fixed bundle in (24). Its fertility objective, after dropping fixed continuation terms, is:

$$\log(c^{y,F} - \chi n) + \alpha \log(h^{y,F} - \kappa n) + \vartheta \log n. \quad (45)$$

At $n = n^F$, its first derivative is zero by (23), and its second derivative is $-\chi^2/(c^{y,F} - \chi n)^2 - \alpha\kappa^2/(h^{y,F} - \kappa n)^2 - \vartheta/n^2 < 0$. It therefore chooses n^F uniquely. Any feasible allocation induced by assigning fixed bundles and anticipating private fertility is feasible in the joint planner problem. The joint optimum is itself induced in this way, so it also maximizes welfare among those assignments.

D Compensation and the local fertility response

Take housing $\epsilon > 0$ from an old household with bundle (c_o, h_o) , and compensate it with goods. Holding its estate fixed, exact indifference requires:

$$\Delta c_o = c_o \left[\left(\frac{h_o}{h_o - \epsilon} \right)^\alpha - 1 \right] = p\epsilon + O(\epsilon^2). \quad (46)$$

The equality uses old optimality, $\alpha c_o/h_o = p$. The receiving parent pays Δc_o and receives ϵ housing. At its original fertility, its first-order utility gain is $(\alpha/s_i - p/x_i)\epsilon > 0$. Small transfers preserve positivity. Applied to positive shares of young and old households, this is a strict improvement in the planner's objective without (A3).

If the parent chooses fertility at its new total bundle, differentiate its private condition. Since $dc = -p dh$ to first order:

$$\frac{dn}{dh} = \frac{\alpha\kappa/s_i^2 - \chi p/x_i^2}{\vartheta/n_i^2 + \chi^2/x_i^2 + \alpha\kappa^2/s_i^2}. \quad (47)$$

The denominator is positive. Substituting $s_i = \alpha v_i/p$ gives (27), completing the local comparison.

E Household policies in all tenure and financing cases

This appendix records the household policies when a tenure or financing constraint binds. These cases include corners excluded by maintained Assumption A1 in the main text; the main theorem is unchanged. All expressions are conditional on prices and on the relevant tenure pair.

E.1 Old households

For an old renter entering with resources z ,

$$h_R^o = \min \left\{ r, \frac{\alpha z}{p(1 + \alpha + \omega_B)} \right\}, \quad c_R^o = \frac{z - ph_R^o}{1 + \omega_B}, \quad e_R = \frac{\omega_B c_R^o}{q}. \quad (48)$$

For an owner, the unconstrained bundle is feasible iff

$$\omega_B \geq \frac{\alpha q(1 - \phi)P}{p}. \quad (49)$$

Then

$$c_O^o = \frac{z}{1 + \alpha + \omega_B}, \quad h_O^o = \frac{\alpha z}{p(1 + \alpha + \omega_B)}, \quad e_O = \frac{\omega_B z}{q(1 + \alpha + \omega_B)}. \quad (50)$$

If the inequality fails, financing binds and

$$c_O^o = \frac{z}{1 + \alpha + \omega_B}, \quad h_O^o = \frac{(\alpha + \omega_B)z}{(1 + \alpha + \omega_B)[p + q(1 - \phi)P]}, \quad e_O = (1 - \phi)Ph_O^o. \quad (51)$$

Evaluate the original old-age utility at the renter bundle and each feasible owner bundle, retaining the larger value. Denote the conditional values by $V_R^o(z)$ and $V_O^o(z)$. The derivatives for retirement choice are

$$V_R^{o'}(z) = \begin{cases} \frac{1 + \alpha + \omega_B}{z}, & z \leq (1 + \alpha + \omega_B)pr/\alpha, \\ \frac{1 + \omega_B}{z - pr}, & z > (1 + \alpha + \omega_B)pr/\alpha, \end{cases} \quad V_O^{o'}(z) = \frac{1 + \alpha + \omega_B}{z}. \quad (52)$$

The owner derivative has this form even when financing binds; the constraint changes the utility level. No blanket unconstrained-old assumption is needed.

E.2 Young households at fixed retirement resources

Fix z and write the current budget as

$$c + ph = w - q(z - y). \quad (53)$$

A renter has $h \leq r$ and $y \leq z < y + w/q$; an owner has $h \leq (z - y)/[(1 - \phi)P]$ and $y < z < y + w/q$. If the unconstrained current home fits the relevant limit, define $x = c - \chi n$ as adult consumption and then

$$x = \frac{w - q(z - y)}{1 + \alpha + \vartheta}, \quad n = \frac{\vartheta x}{\chi + \kappa p}, \quad h = \frac{\alpha x}{p} + \kappa n, \quad c = x + \chi n. \quad (54)$$

Otherwise set h equal to the relevant limit and $c = w - q(z - y) - ph$. Fertility is the unique feasible root of

$$\frac{\vartheta}{n} = \frac{\chi}{c - \chi n} + \frac{\alpha \kappa}{h - \kappa n}, \quad (55)$$

equivalently

$$\chi\kappa(1 + \alpha + \vartheta)n^2 - [\kappa c(\alpha + \vartheta) + \chi h(1 + \vartheta)]n + \vartheta ch = 0. \quad (56)$$

It is the smaller root in $0 < n < \min\{c/\chi, h/\kappa\}$. Some policies therefore require scalar roots; not every case has an elementary closed form.

For each fixed young/old tenure pair, solve the conditional retirement choice separately. With positive saving for a young renter, or slack financing for a young owner, the interior condition is

$$\beta V_d^{o'}(z) = \frac{q}{c - \chi n}, \quad d \in \{R, O\}. \quad (57)$$

A young renter has zero saving, $z = y$, when the following inequality, evaluated at its current optimum with $z = y$, holds:

$$\beta V_d^{o'}(y) \leq \frac{q}{c - \chi n}. \quad (58)$$

For a young owner with binding finance, the condition is

$$\beta V_d^{o'}(z) = \frac{q + p/[(1 - \phi)P]}{c - \chi n} - \frac{\alpha}{(1 - \phi)P(h - \kappa n)}. \quad (59)$$

Each conditional problem has a unique optimum; for a renter it may be the endpoint $z = y$. Compare the original lifetime utilities of the four tenure-pair candidates. Comparing values is necessary because $\max\{V_R^o, V_O^o\}$ need not be concave.